

Question

Can FoldBench PPI support a useful experimental-complex benchmark under a 30% identity threshold?

We compare a strict component-held-out claim with a pair-held-out claim, then freeze a FoldBench-only set released after the inherited Protenix v1 training cutoff.

Two holdout claims

A qualifying alignment has exact identity $\geq 30\%$ over $\geq 50\%$ of the shorter sequence.

Component-held-out rejects a dimer when either chain has any training homolog. Pair-held-out rejects it only when one complex-training document has two chain instances that can be assigned one-to-one to homologs of both candidate chains.

The pair rule tests a new relationship between chain families. It permits prior single-chain exposure.

Survival

The strict rule leaves 0/184 scoped FoldBench dimers and 1/1,929 PINDER dimers after locally auditable MarInFold arms; the last PINDER target is present in Helico fine-tuning.

The pair rule leaves ****35 FoldBench + 183 PINDER = 218**** after both MarInFold complex arms. These include 55 homodimers and 163 heterodimers.

The conservative Helico same-PDB pair screen leaves ****32 + 65 = 97**** candidates.

Confirmation and evidence

The broad complex search hit its 100,000-result median cap. We repeated all 220 initial survivors at a 1,000,000-result cap; its median was 86,868 and it found two additional PINDER exclusions.

Exact MMseqs backtraces provide integer identity counts. Commands, hashes, compact witnesses and final status tables are committed. Complete compressed alignments and logs are published in the public MarInFold HF bucket.

Decision

Use FoldBench under the explicit ****pair-held-out**** claim. Its release-date design protects the Helico comparison from inherited Protenix v1 pretraining exposure. We still remove three targets with a paired homolog in Helico's later fine-tuning pool.

PINDER is not part of the frozen benchmark.

Frozen FoldBench evaluation

The 35 MarFold-clean pairs become **30 natural dimers** after the three Helico fine-tuning exclusions and manual removal of two de novo binder targets.

The set contains 13 homodimers and 17 heterodimers in 26 connected 30%-identity groups. One five-target ubiquitin-related group is kept intact. The deterministic split has **8 development and 22 test targets**; both cuts have the same mean length (448.25 vs 448.27 residues).

Every target ships with its exact FoldBench assembly, canonical two-chain input, label-to-author residue map, resolved cross-chain candidate universe, and contacts-v1 pyconfind ground truth.

Contact-context eligibility

A full-context diagnostic ran 100 rollouts for every target. Seven homodimers had at least one rollout reach the checkpoint's complete 8,192-token context: 147/3,000 rollouts in total. These remain valid structural targets but cannot support the strict exp82 requirement that every rollout finish.

The primary contact and contact-conditioned structural comparison therefore uses **23 context-complete targets** in 19 homology groups. Its metadata-only split has **6 development / 17 test targets**, 6 homodimers and 17 heterodimers, and mean lengths of 361.33 versus 361.29 residues.

Evaluation contract

Contact evaluation ranks only resolved cross-chain residue pairs. R is the number of degree ≥ 0.001 pyconfind interface contacts; no within-chain sequence-separation filter is applied across chains. Contact-budget choices use the 6-target development cut only; the 17-target test cut is read once for the final result.

The structural bundle uses FoldBench's native layout for Helico/DockQ and retains the exact two scored chains even when a deposited assembly has extra copies. Model sample selection must use model confidence; ground-truth best-of-N remains an oracle diagnostic.

The complete 12.3 MB bundle is public at ``hf://buckets/open-athena/MarinFold/data/evals/exp350_foldbench_pair_holdout/v1``.

The context audit and 23-target contact table are public at ``hf://buckets/open-athena/MarinFold/data/evals/exp350_foldbench_pair_holdout/contact_eval_v1``.

Contact result

The exp343 model completed 2,300/2,300 strict rollouts. On the 17-target test split, mean inter-chain R-precision is **0.0260 [0.0036, 0.0453]** versus a random expectation of **0.0028**. The score is about 9.3 times random, though low in absolute terms.

On the six development targets, all-contact top-L gives the best Helico result: mean pair-specific DockQ **0.1284**, versus 0.0381 at L/2 and 0.0329 at L/5. Top-L is frozen before reading structural test results.

Structural result

On 17 test targets, mean DockQ is **0.0362** with contacts withheld, **0.0546** with predicted intra-chain contacts, **0.0697** with all predicted contacts, and **0.8010** with oracle contacts.

The paired all-contact minus withheld delta is **+0.0335 [0.0024, 0.0679]**. All-contact minus intra-only is **+0.0151 [0.0002, 0.0266]**. Both improve 12/17 targets, isolating a positive contribution from predicted inter-chain contacts.

Absolute predicted-contact performance remains low: 2/17 targets reach DockQ ≥ 0.23 , versus 1/17 with contacts withheld and 17/17 with oracle contacts.

Does sampling 1,000 times recover the interface?

Exploratory follow-up: 1,000 attempted rollouts on each of the same 23 targets. Ten unfinished attempts receive zero credit; the target set remains fixed. Individual saved maps exactly reconstruct every aggregate vote.

On 17 test targets, oracle best@100 F1 is **0.110**, rising to **0.192** at best@1000 (matched random: 0.033 and 0.054). Oracle R-precision rises from **0.093** to **0.165**. These require experimental truth to select the sample.

No target has a sample with $F1 \geq 0.5$. The strongest is 8jca: 9 correct contacts out of 29 predicted and 29 true (31% precision and recall).

Real contacts occur, but in different samples

Pooling 1,000 maps recovers **85.7%** of true contacts on average. A matched random pool already recovers **67.2%**, so pooled coverage alone is weak evidence.

The best-of-many F1 advantage over random is real, but the best selected maps average only 26.6% precision and 18.3% recall. Ordinary 1,000-sample consensus reaches **0.0373 R-precision**.

The contact model sometimes emits correct fragments of an interface; these samples do not establish reliable interface recovery. Learned selection or structural realization from this diversity remains untested.

Do simple contact biases explain the sampling advantage?

Two stronger controls use the same saved rollouts. One preserves each map's amino-acid pair counts. The other preserves every individual residue's contact count and shuffles partners through valid bipartite edge swaps.

Test best@1000 F1: **5.4%** for uniform random pairs, **7.3%** for the amino-acid control, **13.5%** for the per-residue-degree control, and **19.2%** for Marifold.

Best@100 F1 is 3.3%, 4.8%, 8.2%, and 11.0%, respectively. These are oracle-selected contact maps, not an inference-time selection method.

Pairing signal survives both controls

Against the per-residue-degree control, the paired best@1000 F1 advantage is ****+5.8** percentage points [3.9, 8.8]**, positive on 16/17 test targets. R-precision is 16.5% versus 11.3%.

The degree control preserves any learned localization of interface residues; it tests the additional pairing information. Its score is not solely an amino-acid-size effect.

The sampler is approximate. Five times more shuffling yields 13.50% F1 versus 13.48%. Small-graph exact enumeration, degree/duplicate checks and source hashes validate the implementation. Absolute quality remains limited: no original sample reaches F1 50%.

Contact-map atlas: usually wrong, sometimes absent

The 16-page `contact_prediction_atlas.pdf` compares experimental contacts with frequency heatmaps, top-R errors, typical samples and oracle samples for six contrasting test complexes. Cohort summaries include all 17 targets.

Of 17,000 attempts: **20.3% emit no inter-chain contacts**, **1.4% emit only unscorable pairs**, **61.6% emit scorable contacts but get none correct**, and **16.7% include at least one correct pair**. Eight attempts are unfinished.

Failure patterns differ: 8cqm's selected contact endpoints miss both interface regions; 7ytu samples 35/37 true pairs but ranks none in top-R; 8smq often selects real interface residues but pairs them incorrectly. The atlas preserves missing-residue masks and shows false positives in every selected map.

Four models, the same complex predictions

All four arms predict each complete dimer. Intra-chain scores use the two original partners from those same predictions; no monomer-only evaluation targets are used. The headline remains the fixed 17-target test split (13 homology groups).

Test intra / inter R-precision: Marifold multichain ****44.3% / 2.6%****; current default + ten glycines ****45.1% / 1.4%****; ESMFold2 ****74.8% / 43.9%****; AlphaFold3 ****76.7% / 52.9%****.

Marifold uses 100 rollouts per model. ESMFold2 uses native dimers without MSAs; AlphaFold3 uses ColabFold paired/unpaired MSAs and no templates. Both structure models select one of five samples using model confidence. Their training-data holdout has not been separately audited here.

Interface failure survives the linker control

Joining the partners with ten glycines preserves similar within-chain accuracy but does not rescue the interface. The multichain model is stronger at inter-chain contacts than the linker control in point estimates, but both remain far behind the structure baselines.

Ground truth and predicted structures use the same pyconfind contact operator. The native resolved mask is identical for all models. Intra-chain R pools contacts from both partners; separate per-chain and long-range scores are also saved. Ten-glycine linker residues are excluded from every score.

The dedicated five-page `four_model_comparison.pdf` contains the aggregate plot, per-complex heatmaps, intra-versus-inter scatter plots, and per-partner accuracy. All plots use the same 17 test complexes. Error bars bootstrap the 13 homology groups.

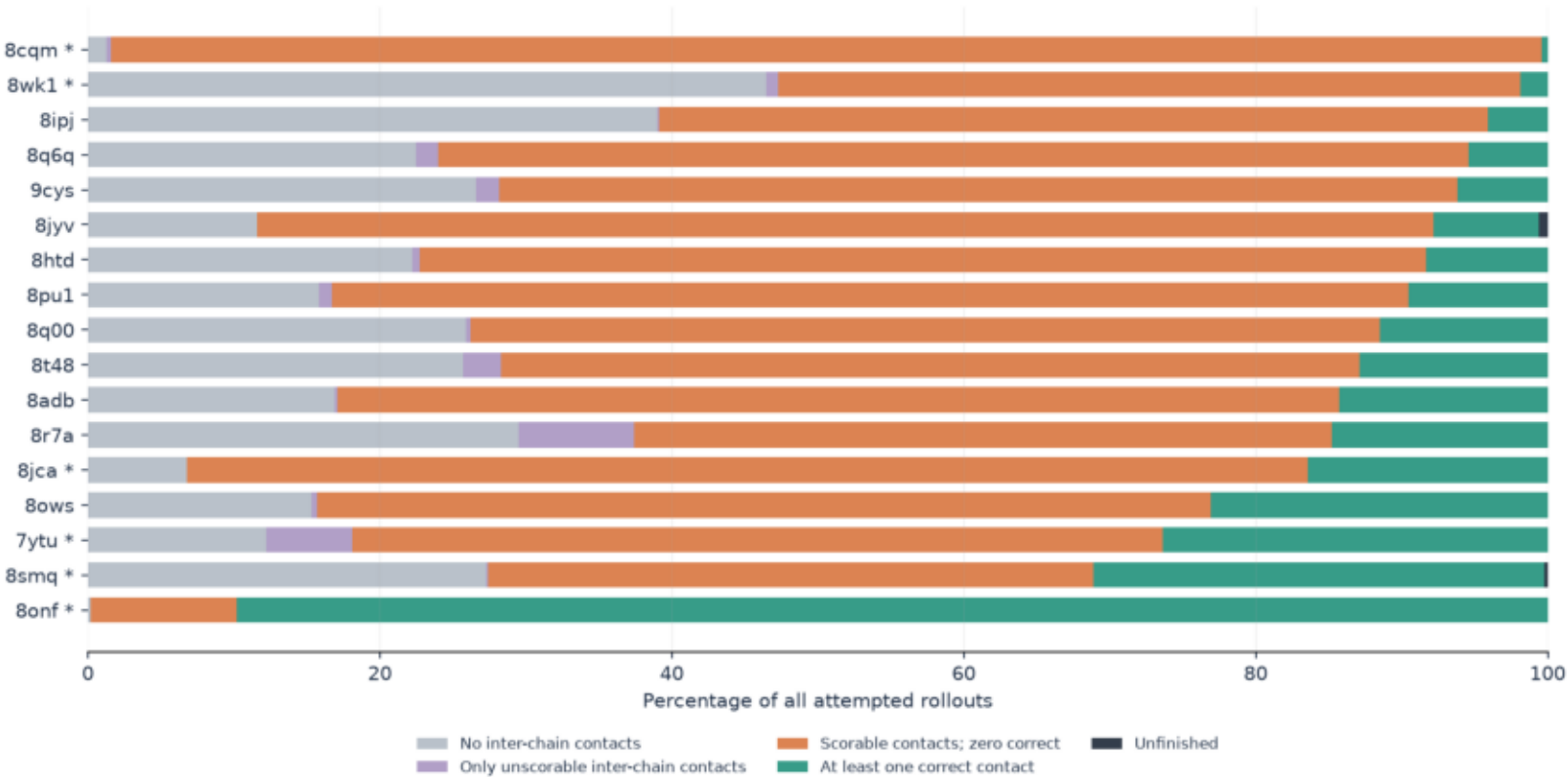
contact_failure_modes.png

All 17 test complexes, categorized by raw inter-chain emissions, resolved scoring and experimental overlap.

How often is a rollout empty, wrong, or partly right?

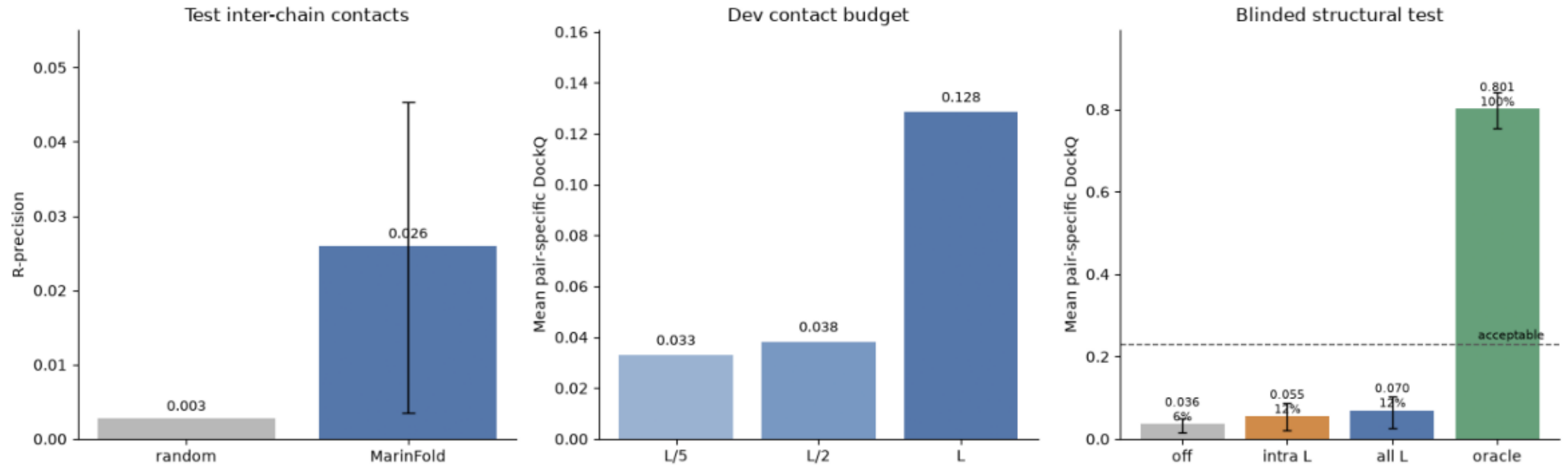
Each bar represents 1,000 attempts. A partly right map contains at least one true contact; it can contain many false contacts.

* Included in the detailed case studies



eval_results.png

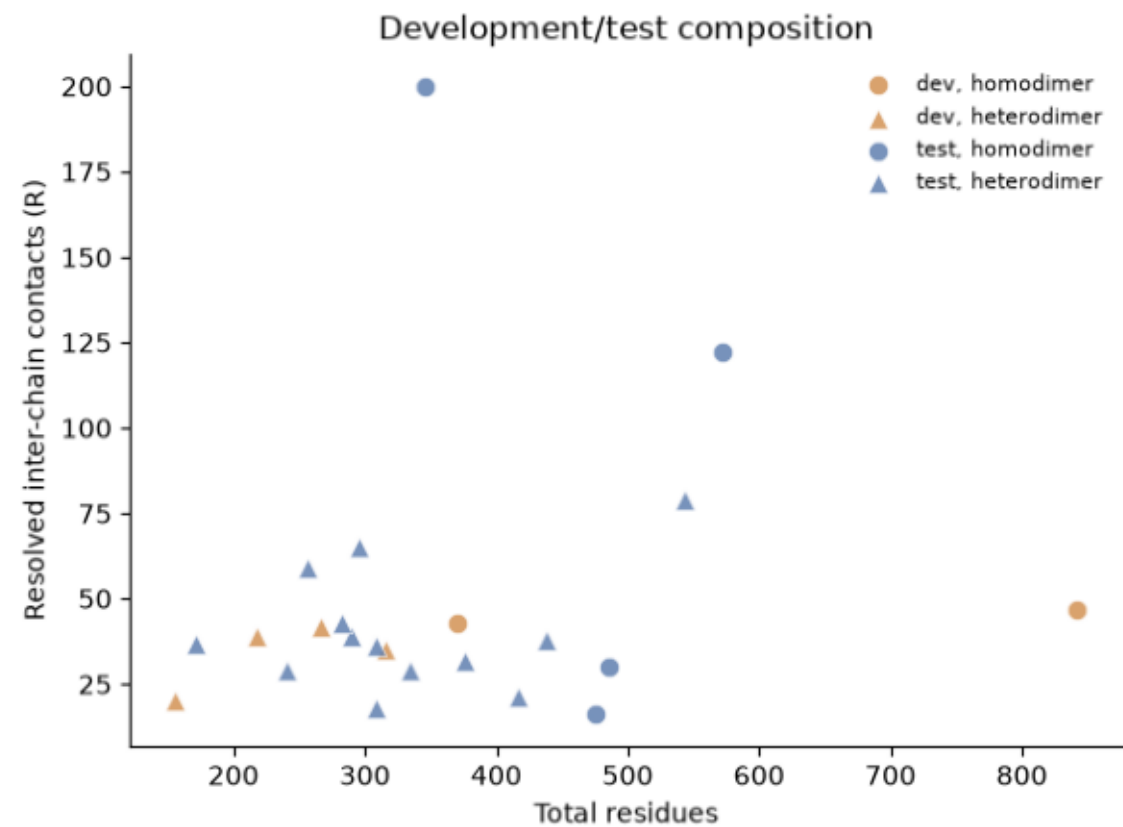
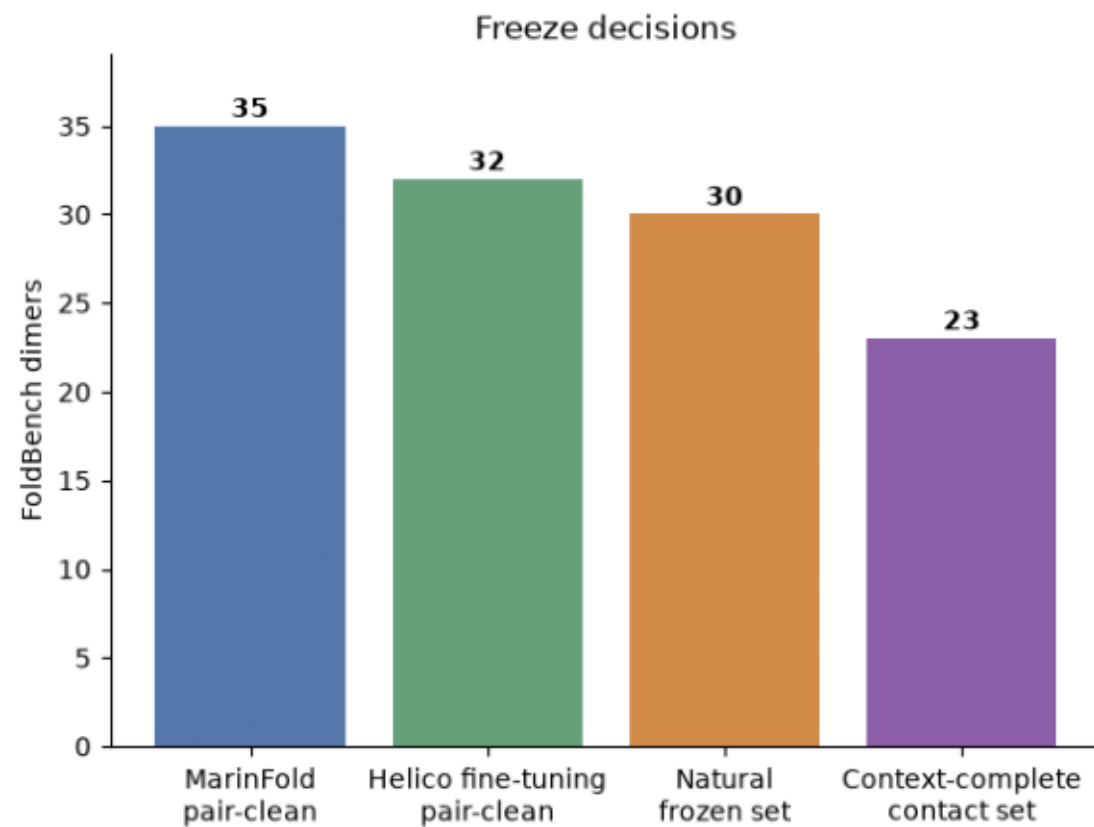
The contact model beats the resolved-pair random baseline on the 17-target test split. Helico contact budgets are selected only on the six-target development split; the test bars use confidence-selected samples and exact pair-specific, symmetry-aware DockQ. Error bars are 95% homology-group bootstrap intervals.



```
$ plot_eval_results.py
```

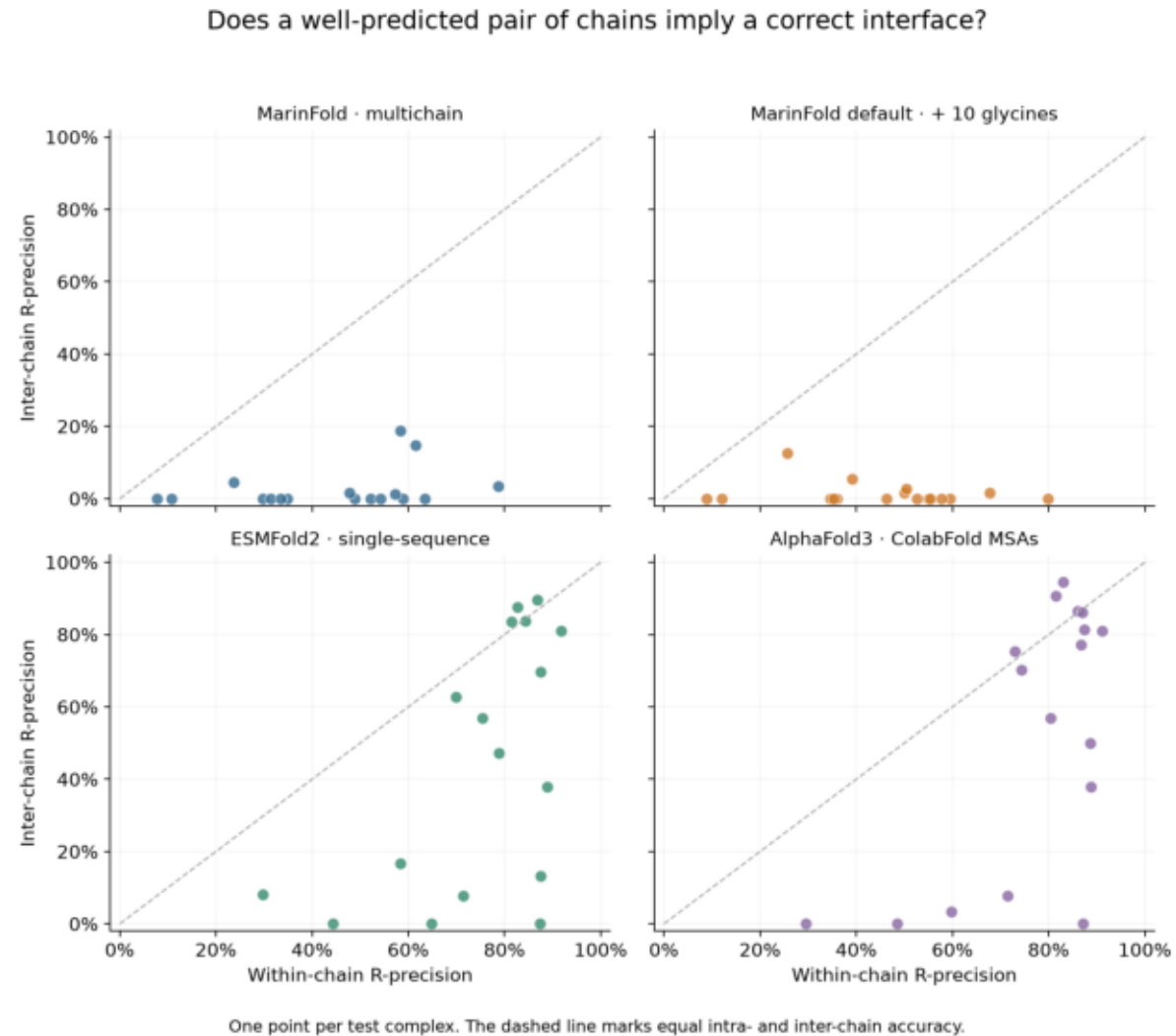
foldbench_eval_freeze.png

The 35 Marifold pair-clean FoldBench dimers yield 30 natural structural targets, of which 23 complete all 100 contact rollouts inside the model's 8,192-token context. The 6/17 contact split is grouped by chain homology and balances mean total length.



four_model_intra_inter_scatter.png

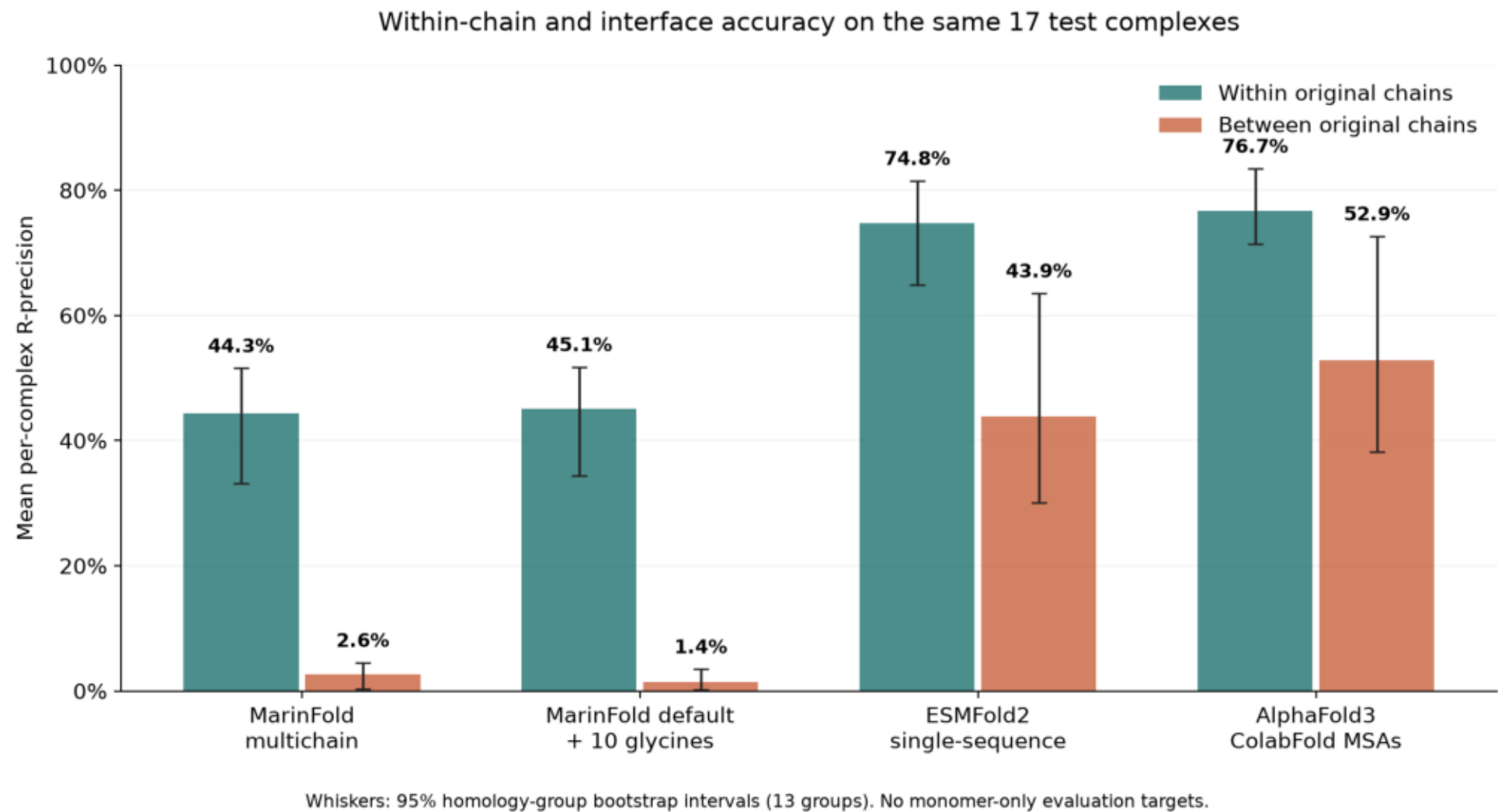
Paired within-chain and inter-chain accuracy for each full-complex prediction; common axes across all models.



```
$ plot_four_model_eval.py
```

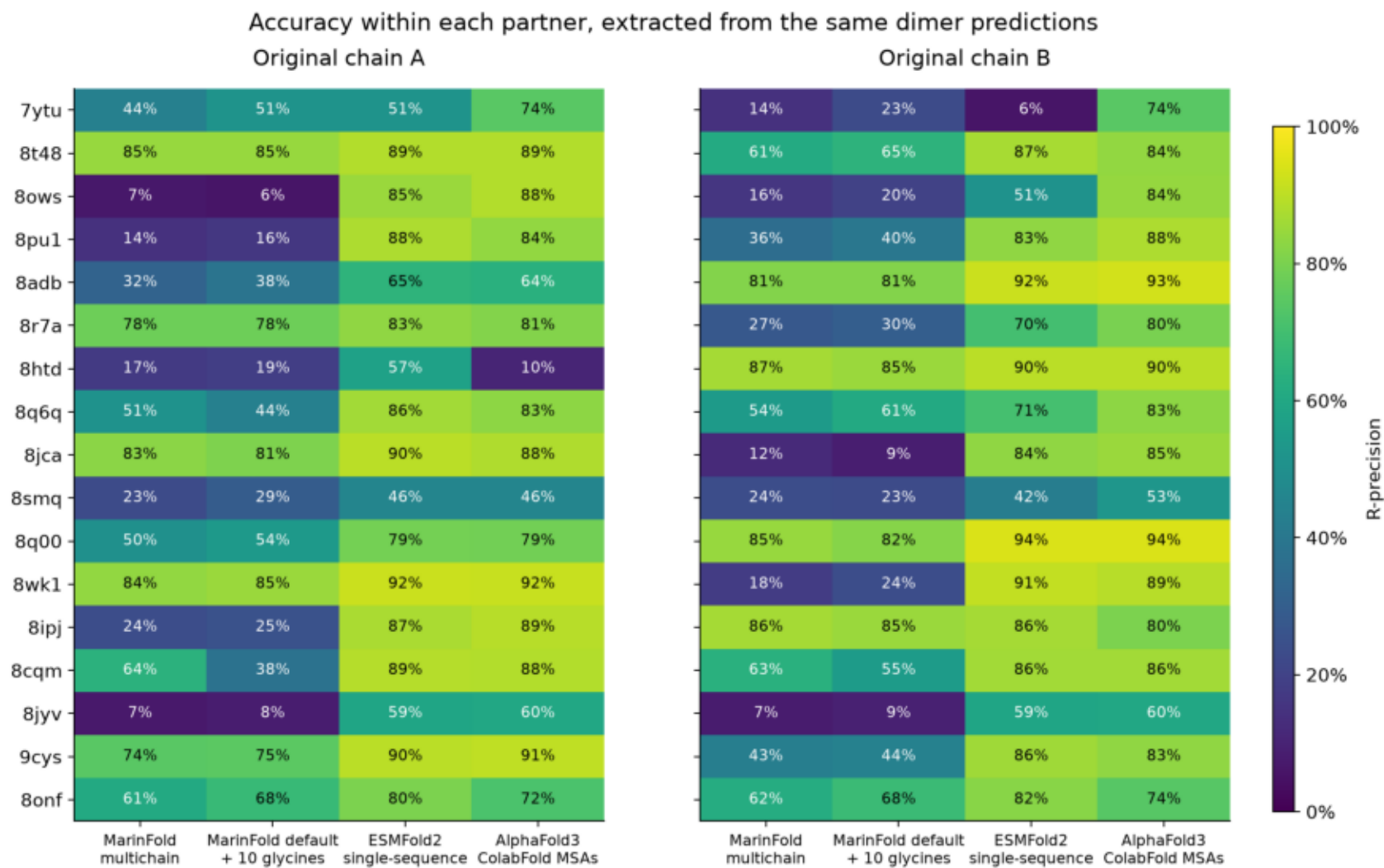
four_model_overview.png

Same 17 test dimers in both contact classes and all four model arms; means and 95% group-bootstrap intervals.



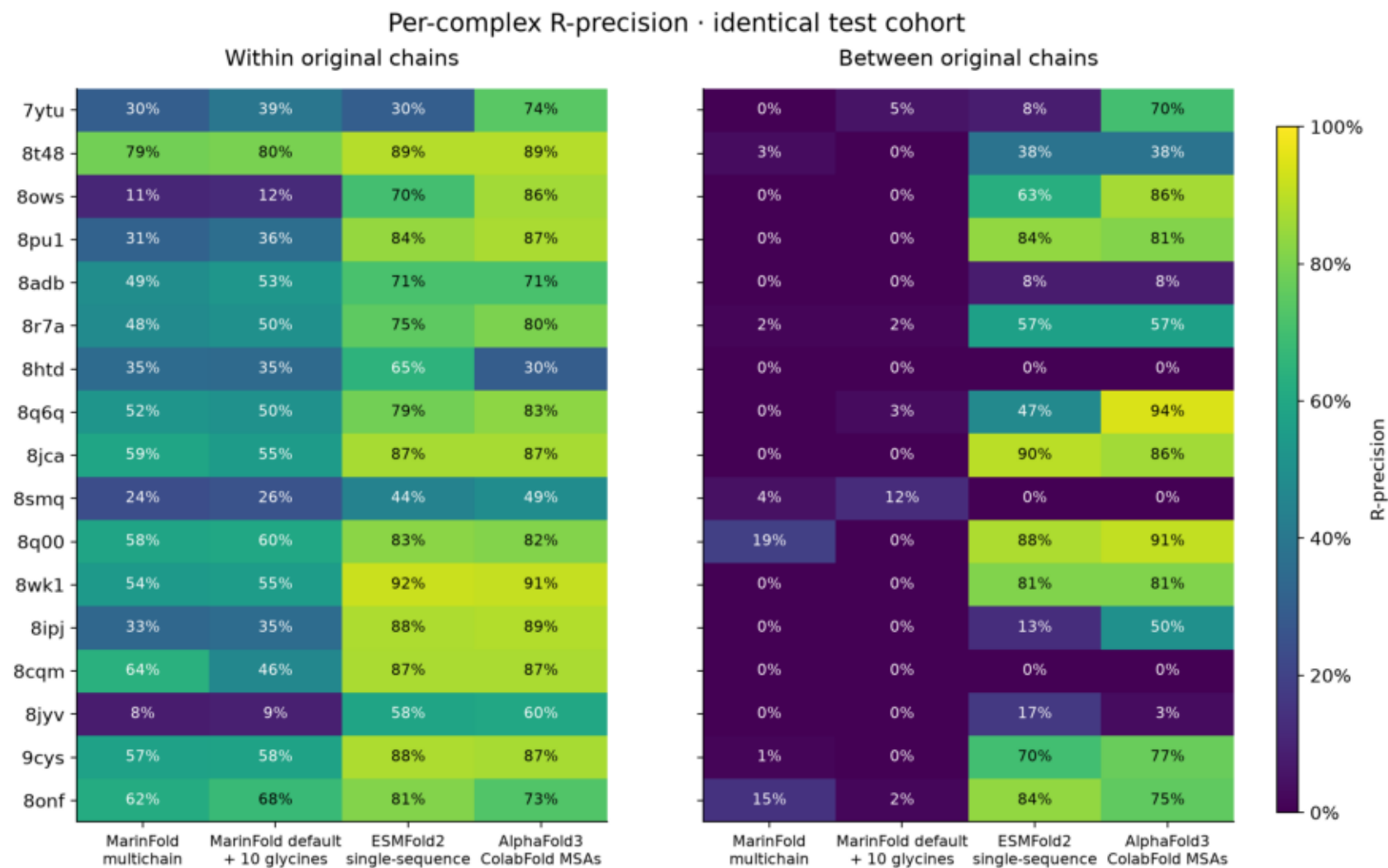
four_model_per_chain.png

Per-partner R-precision using each chain's own number of true contacts; chain labels follow the frozen input order.



four_model_per_complex.png

Every test complex, ordered by total sequence length; all cells share a 0–100% color scale.



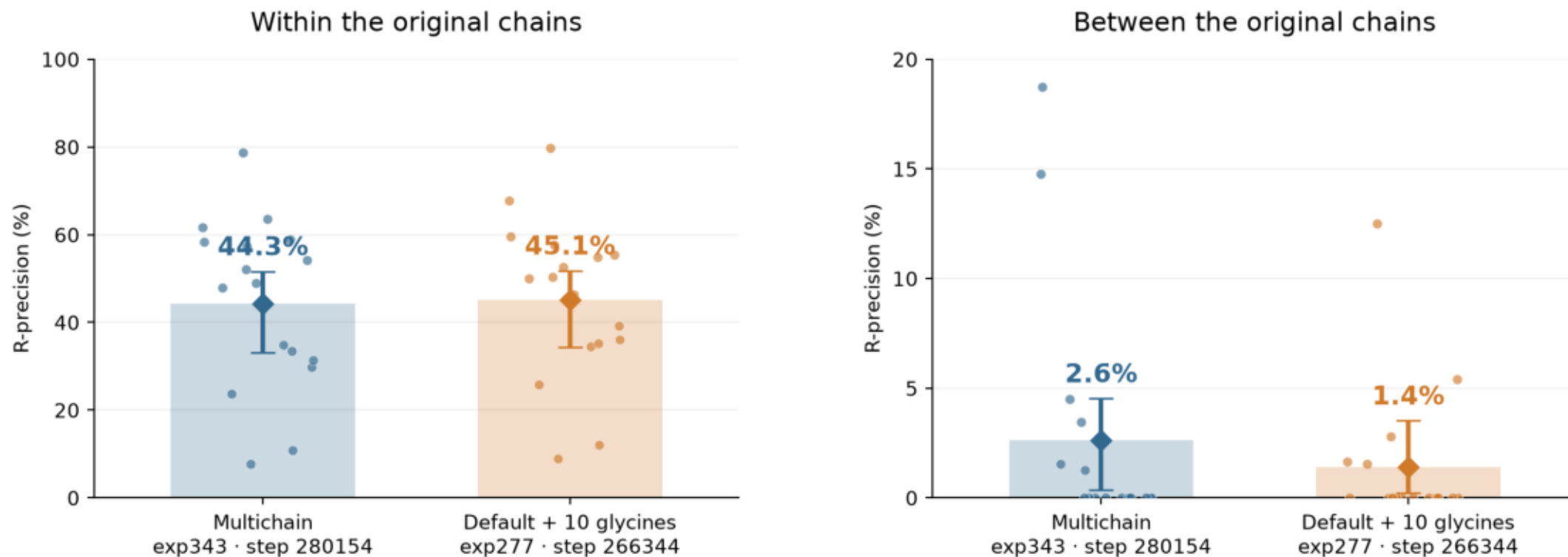
\$ plot_four_model_eval.py

four_model_progress.png

Interim comparison of the two completed MarinFold arms on the same 17 complex test targets; intra-chain scores use both original partners. 95% homology-group bootstrap intervals. Inter-chain axis is enlarged 5x.

MarinFold predicts contacts within chains much better than between them

Same 17 held-out test complexes · 100 rollouts per model · native resolved residues only

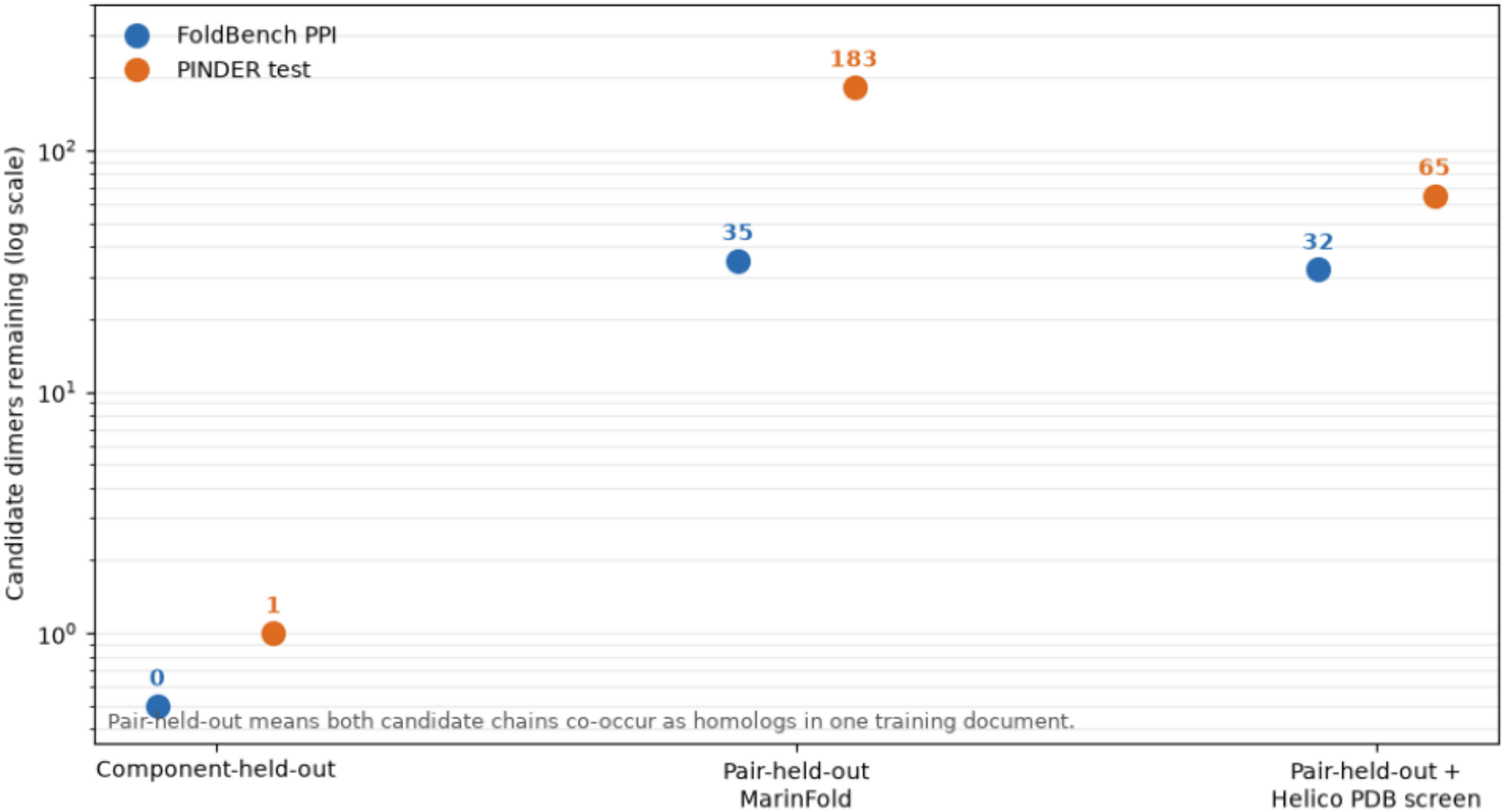


Dots: individual complexes. Diamonds: mean. Whiskers: 95% bootstrap interval over 13 homology groups.

Different y-axis scales: the inter-chain panel is enlarged 5x. See four_model_comparison.pdf for all four models.

pair_survival.png

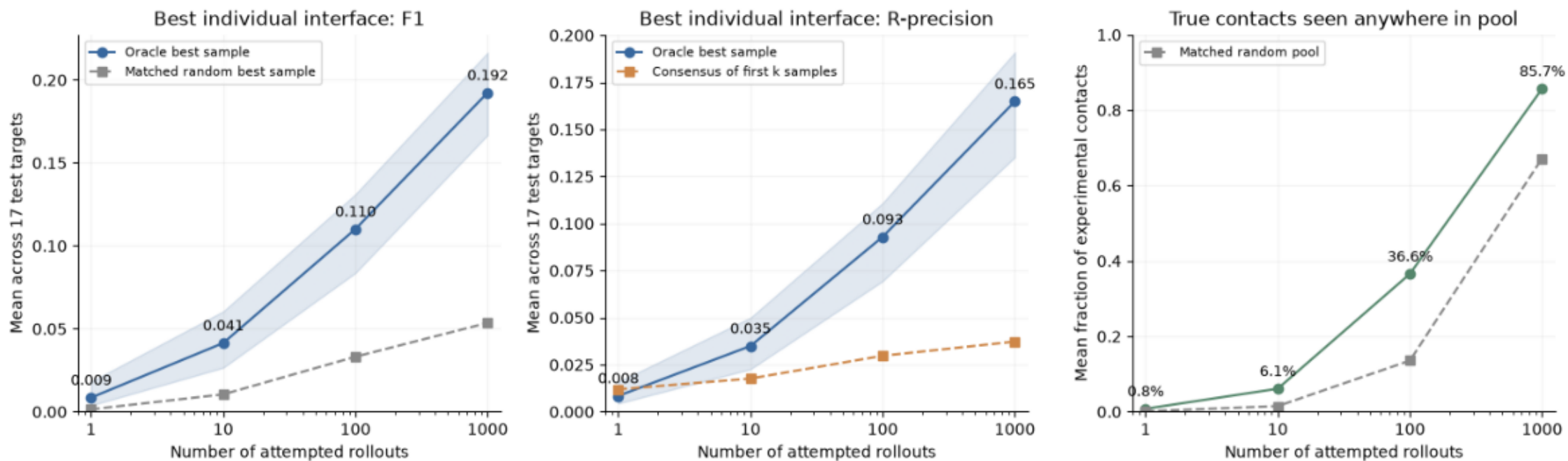
The strict component-held-out rule rejects a dimer when either chain has a training homolog. The pair-held-out rule rejects it only when one complex training document contains one-to-one homologs for both candidate chains. The final column additionally applies a conservative same-PDB pair screen against Helico's documented fine-tuning pool.



rollout_sampling.png

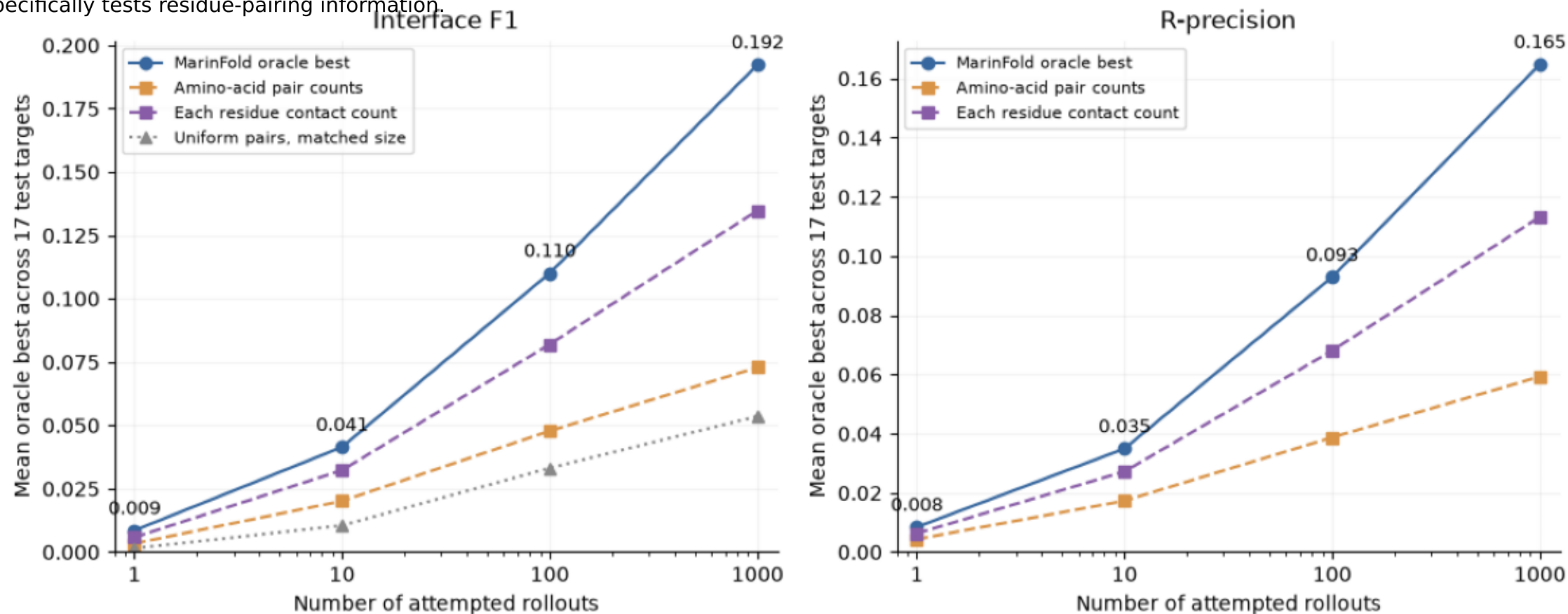
Exploratory follow-up on the frozen 17-target test split, 1000 attempted rollouts per target. Oracle curves average exact best-of-k expectations over subsets of each saved pool; shading is the 95% homology-group bootstrap interval. F1 and R-precision are maximized separately; single-map R-precision averages tied binary scores. Consensus uses fixed prefixes and the original stable top-R convention. The matched random F1 control preserves each sample contact count. Union recall pools contacts across samples and does not imply a single correct map. Unfinished attempts receive zero credit.

MarinFold inter-chain sampling: oracle diagnostic on pair-held-out FoldBench



sampling_controls.png

Same 17 frozen test targets and 1000 attempted maps per target. All curves use ground-truth oracle selection; F1 and R-precision are maximized separately. The AA control exactly preserves each chain-A/chain-B residue-type pair count. The degree control preserves every residue degree using a uniform-stationary bipartite switch chain (100 proposals per edge burn-in, 20 between draws; 1000 null pools). The latter is approximate and checked against longer chains, as recorded in the result tables. Ten unfinished attempts retain zero credit. Degree conditioning also preserves any genuine interface-residue localization, so it specifically tests residue-pairing information.



survival.png

Natural protein-only dimers after scope/quality filters and cumulative sequence exclusion against four locally available exp343 training arms. Every constituent chain must avoid a $\geq 30\%$ identity alignment covering $\geq 50\%$ of the shorter sequence. The remaining PINDER target is an upper bound because the ProteinMPNN redesign corpora were not searched.

